Zilong Zhang

Ph.D. Student in Statistics and Machine Learning, Georgia State University
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EDUCATION

Ph.D. in Machine Learning & Data Science **Aug. 2025 – Present**
Department of Mathematics & Statistics / Georgia State University *Atlanta, GA*

M.S. in Statistics **Aug. 2023 – May 2025**
Department of Statistics & Probability / Michigan State University *East Lansing, MI*

Bachelor of Financial Mathematics **Sep. 2019 – Jul. 2023**
School of Statistics / University of International Business and Economics *Beijing, China*

RESEARCH PROFILE

- Statistics and machine-learning Ph.D. researcher studying how to make model evaluation more reliable when human labels are scarce, selective, or costly to obtain.
- Research interests include LLM-as-a-judge auditing, human-aligned evaluation, positive–unlabeled learning, uncertainty-guided review, and graph/sequence models for biomedical data.

PUBLICATIONS & MANUSCRIPTS

- **Zilong Zhang**, Yi-Ting Hung, Weiyi He, Junxi Zhang, Lei Ding, Chi-Kuang Yeh. "AURA: Adaptive Uncertainty-aware Refinement for LLM-as-a-Judge Auditing" *Under review, arXiv:2606.19714*, 2026.
- **Zilong Zhang**, Yi-Ting Hung, Lei Ding, Chi-Kuang Yeh. "Quantifying and Auditing LLM Evaluation via Positive–Unlabeled Learning" *Under review, arXiv:2606.19057*, **2026**.

RESEARCH AND OTHER PROJECTS

Uncertainty-Aware Auditing of LLM-as-a-Judge Systems **2026**

- Developed an adaptive auditing approach for pairwise LLM judgments that estimates alignment with human preferences, propagates reliable evidence, and prioritizes uncertain comparisons for limited human review. Evaluated on synthetic and real pairwise-response data.
- Complementary work frames selectively human-labeled comparisons as positive–unlabeled learning and uses partial optimal transport to align verified positives with a reliable subset of unlabeled examples, improving human-preference alignment and robustness to judge and verbosity bias without retraining the judge.

Functional Time-Series Research **Sept. 2025 – Present**

- Research on bandwidth selection for functional time series and long-run covariance estimation.

mRNA-to-Protein Prediction with Transformers and GNNs **Jan. 2024 – Jan. 2026**

- Built a Transformer/GNN model to predict protein expression from single-cell RNA data, using complementary sequence/context and graph representations.
- Conducted the cross-institution project with collaborators at Michigan State University and MD Anderson Cancer Center, connecting single-cell RNA inputs to protein-level outcomes.

WORK EXPERIENCE

China Securities Co., Ltd. **Jul. 2021 – Nov. 2021**
Investment Management Intern *Beijing, China*

- Prepared industry research, risk assessments, listing materials, and fundraising reports for client companies.

TECHNICAL SKILLS

Programming: Python, R, C; PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers
Research methods: LLM evaluation, positive-unlabeled learning, partial optimal transport, uncertainty estimation, GNNs
Tools: Git, Linux, Jupyter, VS Code